

Cross-talker generalization: figures and analysis update

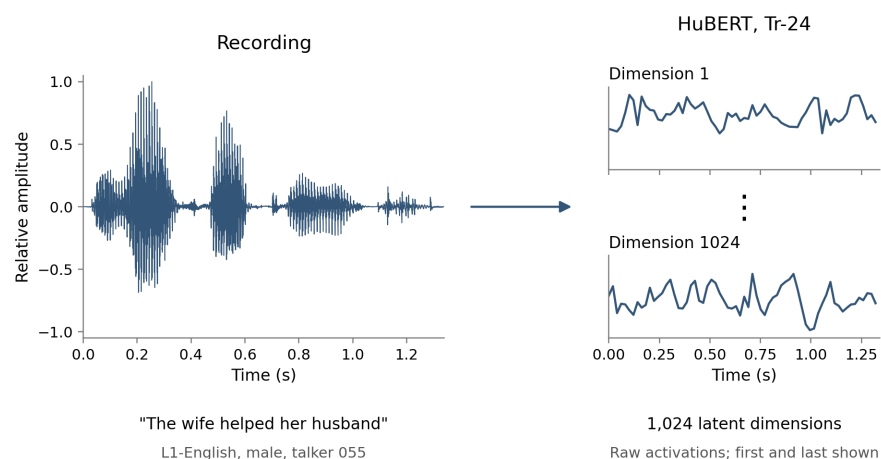
I have revised the figures following your PDF annotations and the messages about Figure 1 and the AN19 phoneme analysis. The main changes are:

1. **One behavioral ceiling.** You are right that the ceiling does not depend on SBI versus HVE. My earlier wording was misleading: the old ceiling tables are identical. What I need to match is the response sample, folds, statistic and fitting procedure.
2. **Shared figure style.** I have adopted the language groups and colors in your world-map code. I have added both distance and similarity matrices with English talkers, marginal distributions below the control curves, and the black pooled X21 curves.
3. **Layerwise likelihood.** I recovered the three-fold predictor-only scores for all 18 registered HuBERT layers and both acoustic baselines. The new likelihood plots complement the z plots. They come from retained runs, not a new parameter search.
4. **The X21 HVE comparison.** Condition does add to HVE: $p = .0117$. HVE itself has a strong negative association before condition is added. The $p = .053$ result concerns HVE's additional contribution after controlling for condition, not its predictor-only association.
5. **Remaining analyses.** I need to rerun the acoustic components with matching scaling and make the direct SBI/HVE comparisons use matching data and model structures. For the AN19 phoneme panels, I propose evaluating an alignment pilot rather than treating missing labels as a permanent barrier.

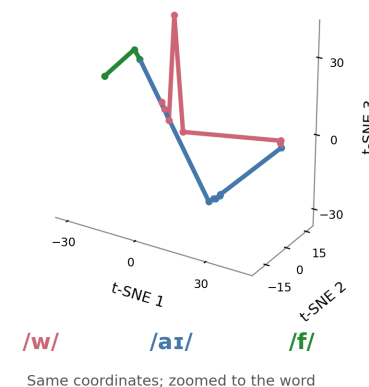
I explain these points below. The complete figure set and source tables accompany this report.

Figure 1a-b: simplifying the representation example

a Speech to latent representation



b Segments in the word "wife"

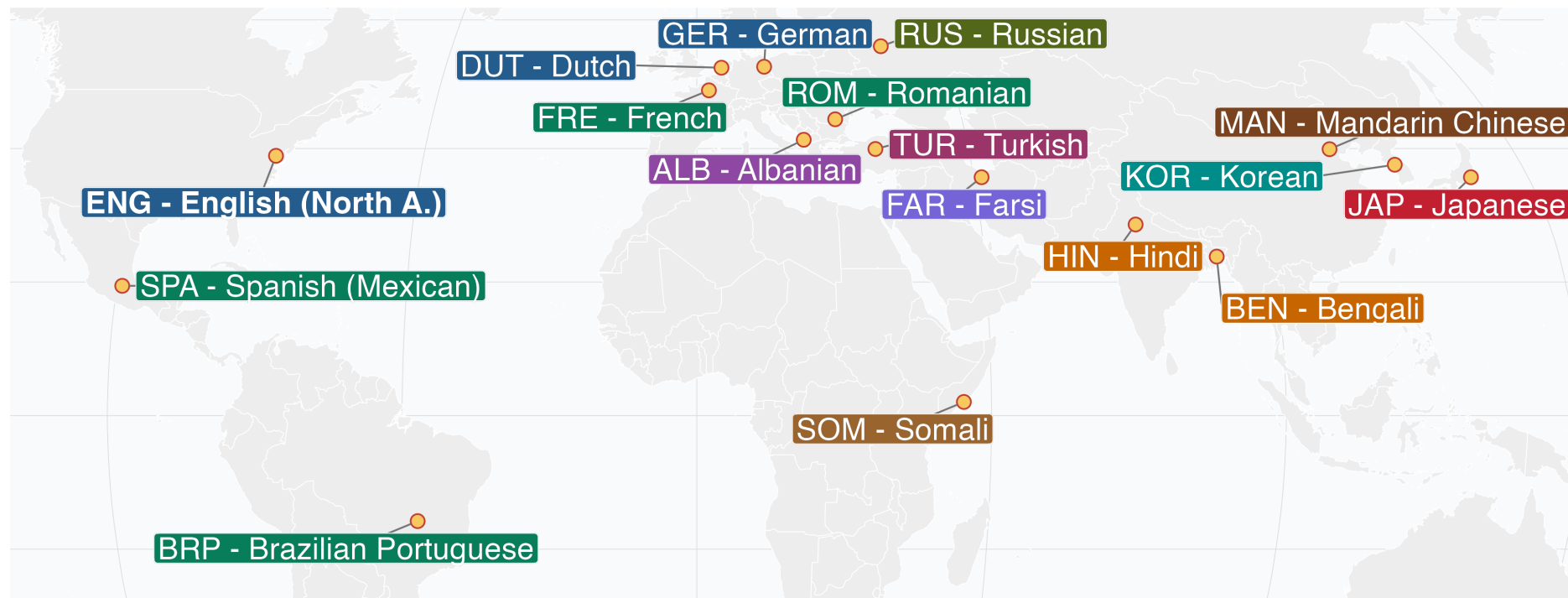


I have grouped the waveform, latent dimensions and t-SNE trajectory into panel a, with the word/segment illustration in panel b. I show Dimension 1 and Dimension 1024 with a vertical ellipsis between their axes, use a white background, and keep the word and phone labels separate from the trajectories.

I changed the example to "The wife helped her husband", from the same English-L1 male talker 055. All five English manifests label the earlier sentence "A boy fell from a window", so I have not changed its article without verifying the recording. The new example uses the existing Tr-24 features and annotated word "wife" (W, AY1, F); it also avoids the previous one-frame vowel. No features or t-SNE coordinates were recomputed.

Figure 1c: language backgrounds

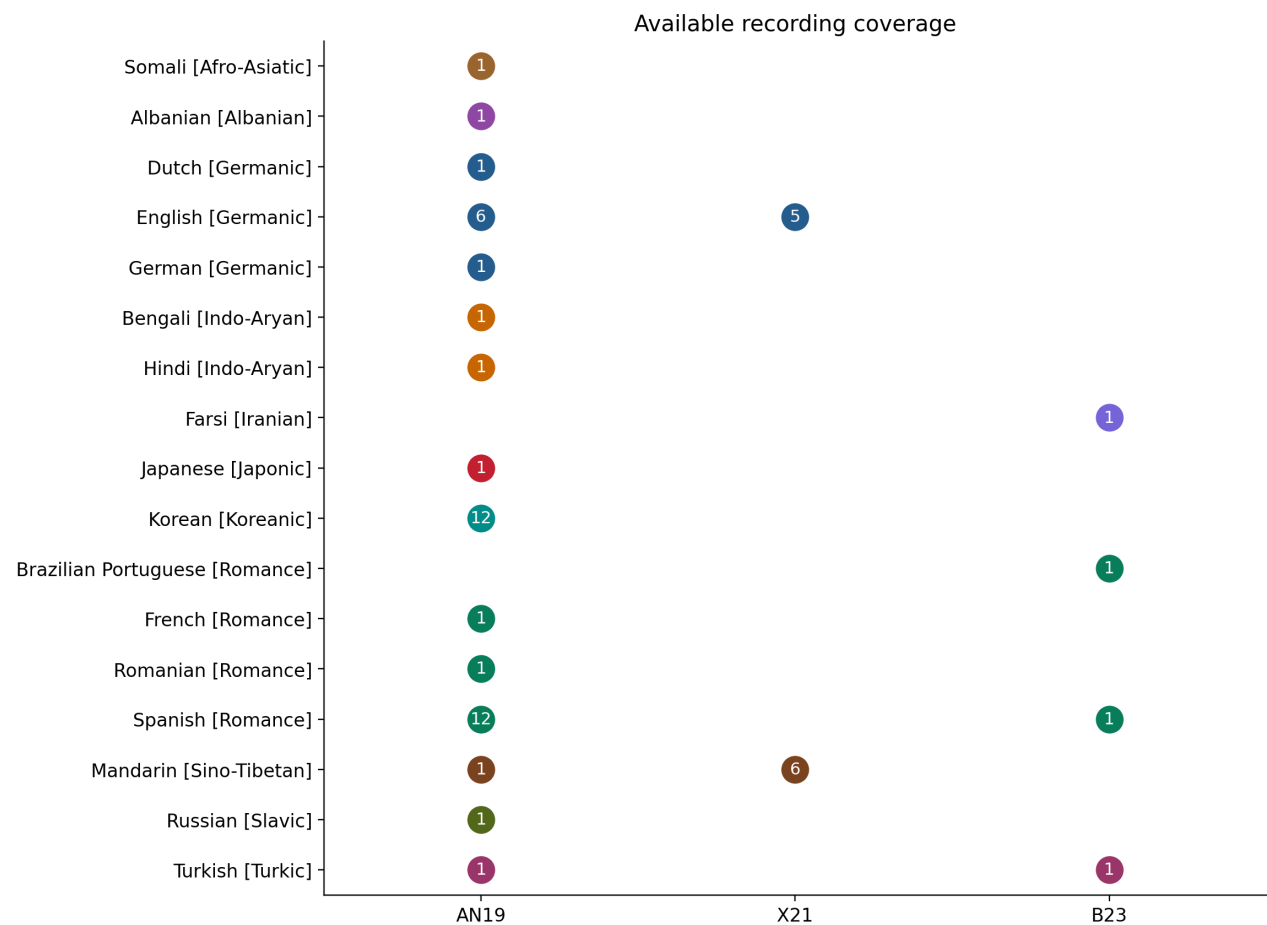
First language backgrounds represented in the data



I have retained your map and used the exact language-group colors from its R script in the language coverage, inventory and control-intelligibility figures. I have preserved which languages share a color.

The locations are the associated national capitals used in your script, not the individual talkers' recording locations. The coverage table on the next page is the companion figure for the supplementary information.

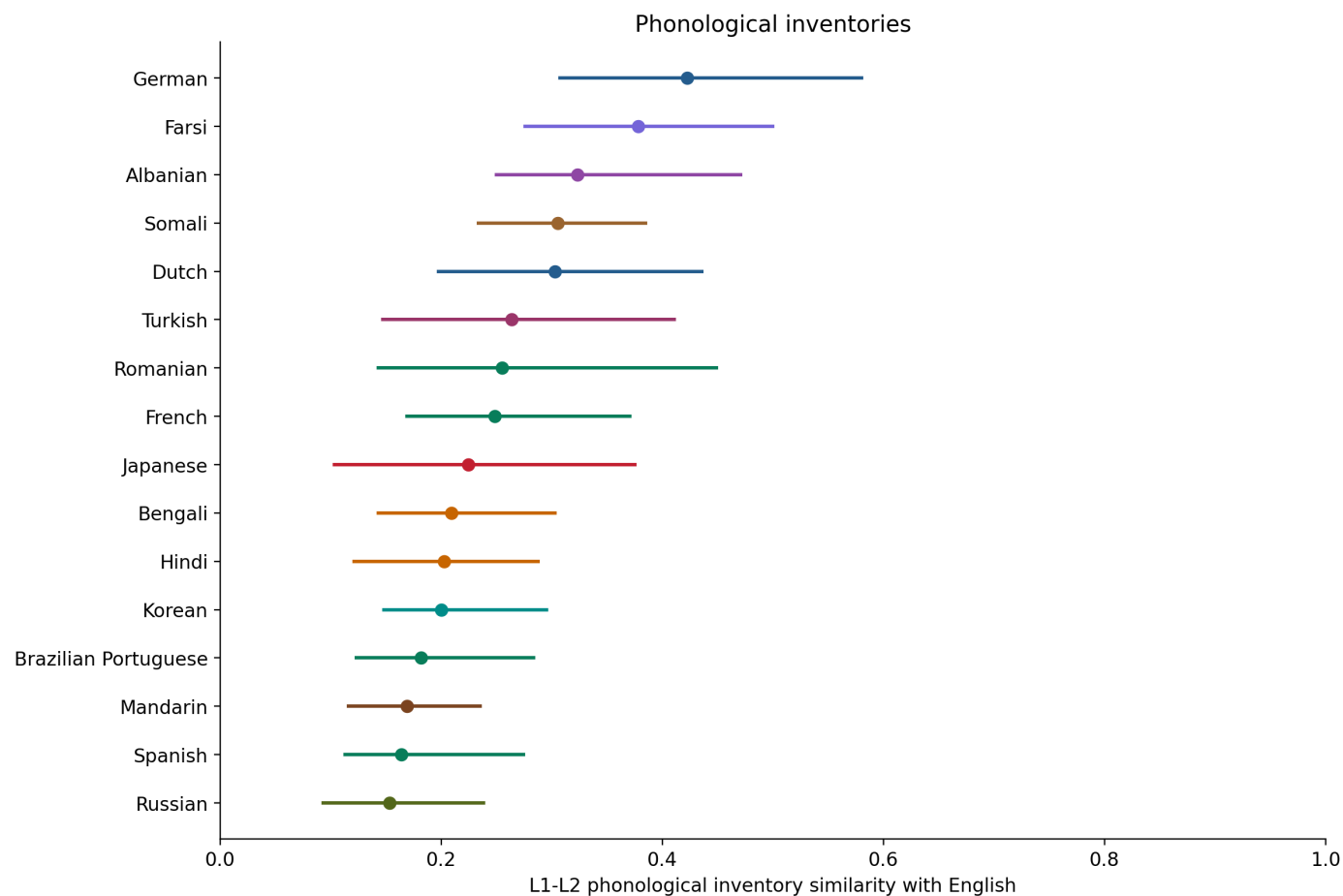
Supplementary figure: talker coverage by dataset



I have kept this figure for the supplementary information, as you suggested. Counts refer to the available speech corpora, including the English reference recordings, rather than listeners or exposure conditions. There are 17 L1 labels across the datasets.

I use Hindi and Mandarin for the original AN19 labels "Indian" and "Chinese", following the study's Table I. The color groups now follow your world map, including its Afro-Asiatic and Sino-Tibetan group labels.

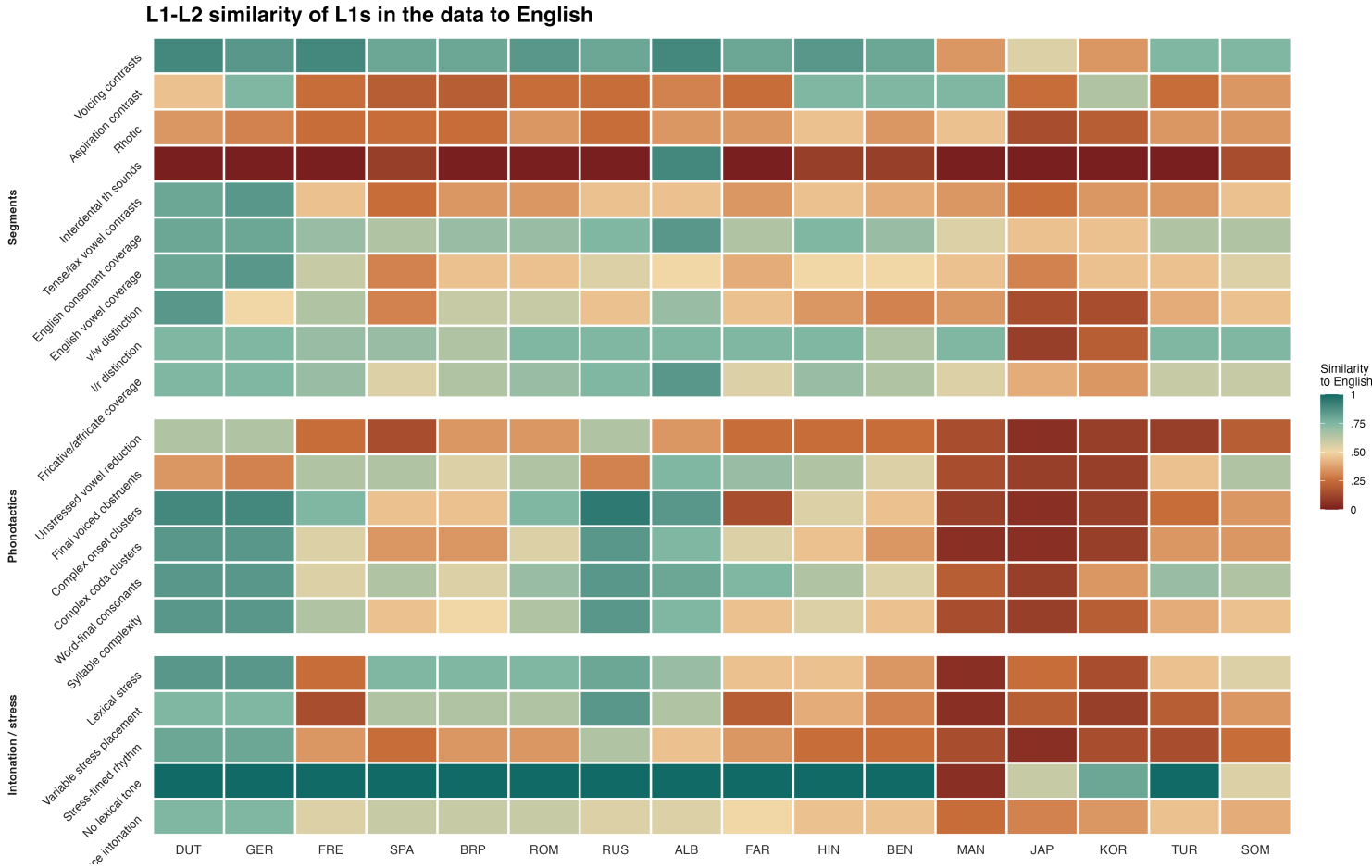
Figure 1d: inventory similarity to English



I removed inventory counts, updated the axis wording and matched the map colors. Points show mean consonant/vowel inventory Jaccard overlap with English; lines show source ranges, not confidence intervals.

I would place this summary above your feature examples. Before adding tone, I need to distinguish absent tone from missing annotation and specify its weighting. Source: Moran & McCloy (eds.), [PHOIBLE 2.0](#), v2.0, CC BY-SA 3.0.

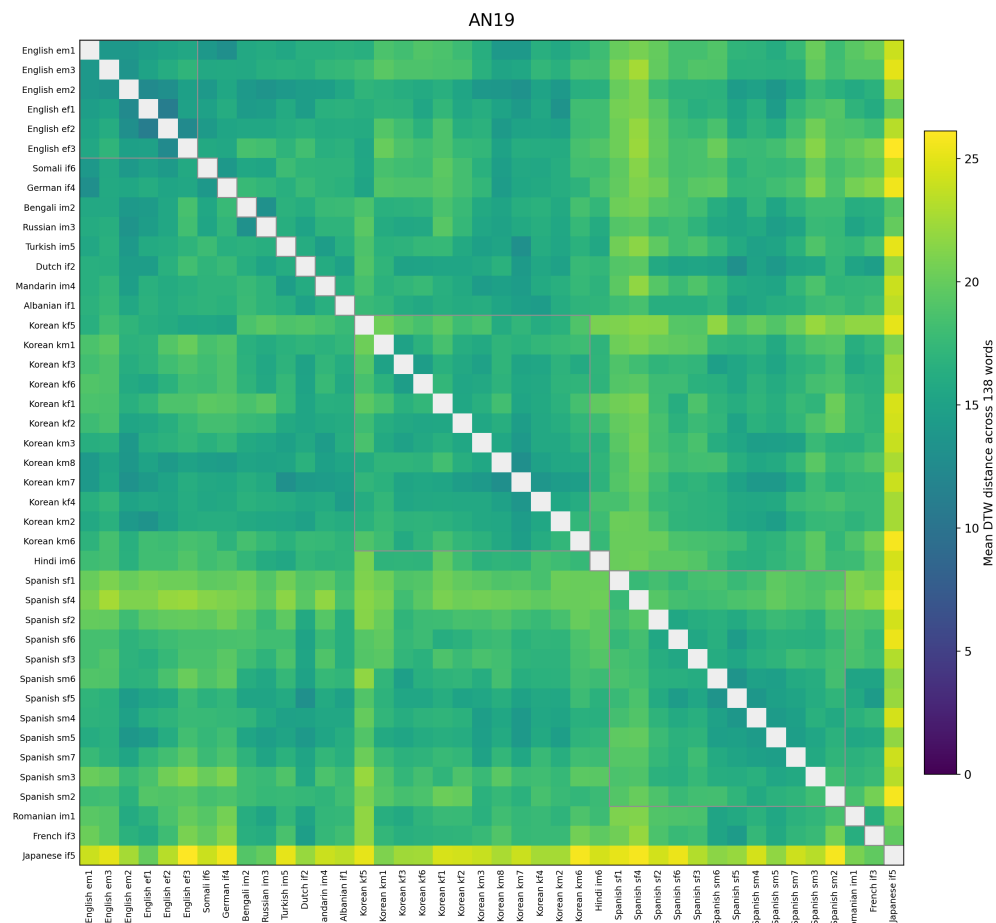
Figure 1d companion: specific phonological features



I have included your feature comparison alongside the inventory summary. It covers segmental, phonotactic and prosodic properties that the consonant/vowel inventory calculation does not capture.

I have not recomputed these cell values or treated them as PHOIBLE-derived measurements. I have the supplied image/PDF and the map code, but still need the heatmap's score table and generation code to document its scale and reproduce it. The last row label is clipped in the supplied graphic; I would fix that in its source before combining the panels for the paper.

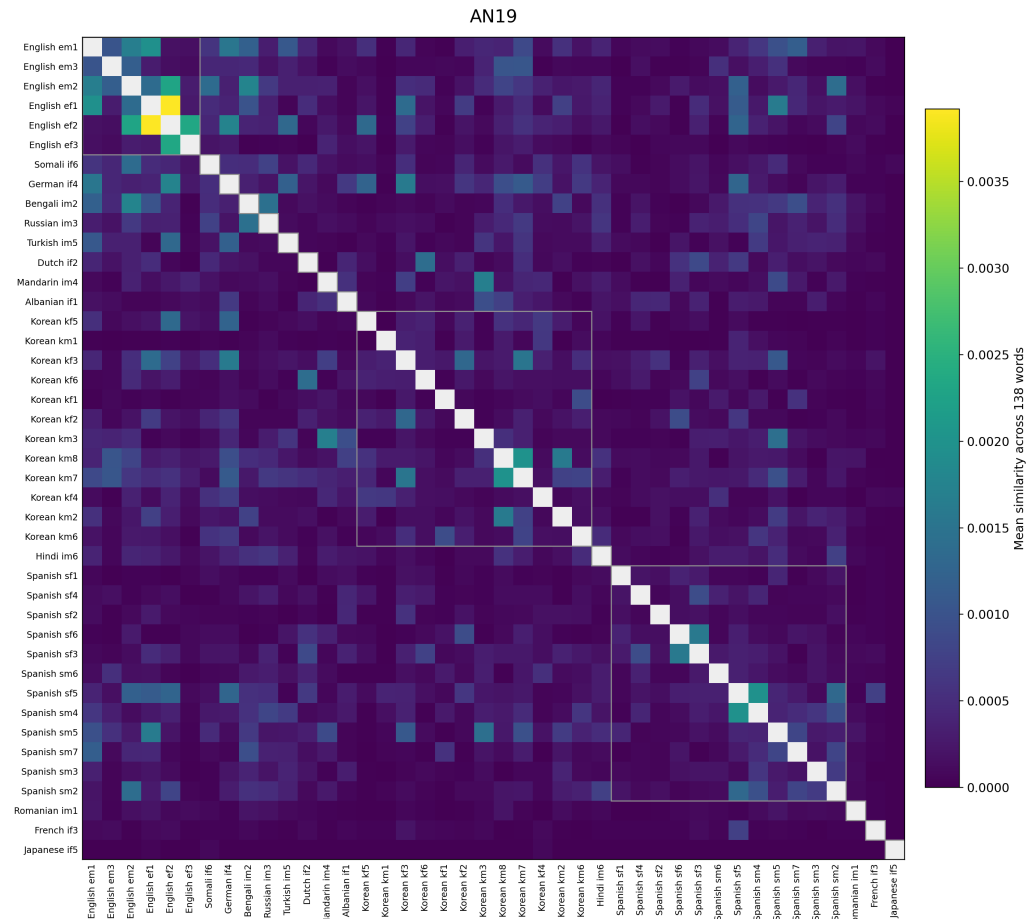
Figure 2a: mean pronunciation distance



I have added the six English reference talkers, bringing the matrix to 42 talkers. Each cell averages same-word DTW distance across the same 138 words. Labels begin with the language and are black; gray outlines identify within-language blocks.

English appears first. I ordered the other language groups by mean distance to English and clustered talkers within each language using their pairwise distances. This ordering is descriptive and uses no intelligibility outcomes. The color scale is linear, so darker/lighter differences are not expanded by a log transformation.

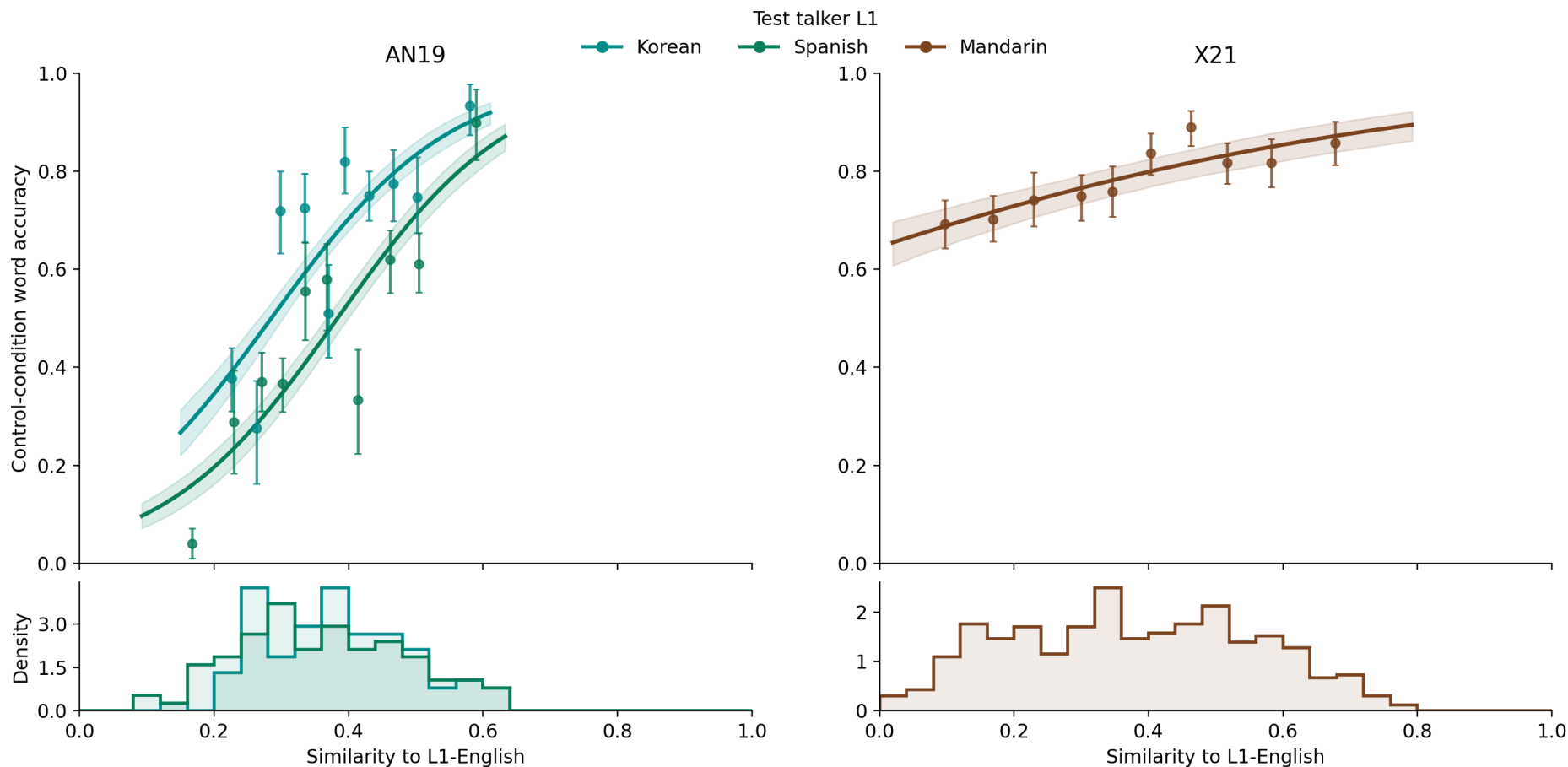
Figure 2a companion: mean pronunciation similarity



This is the similarity version with the same talker order and a linear color scale. I retained $k = 1$: each cell is the mean of $\exp(-\text{DTW distance})$ across 138 words. It is not $\exp(-\text{mean distance})$.

The two panels therefore show different summaries, not just different labels for the same colors. I removed the logarithmic display and shortened the colorbar label as requested. Self-comparisons remain masked in both panels.

Figure 2c: similarity to English and control-test accuracy

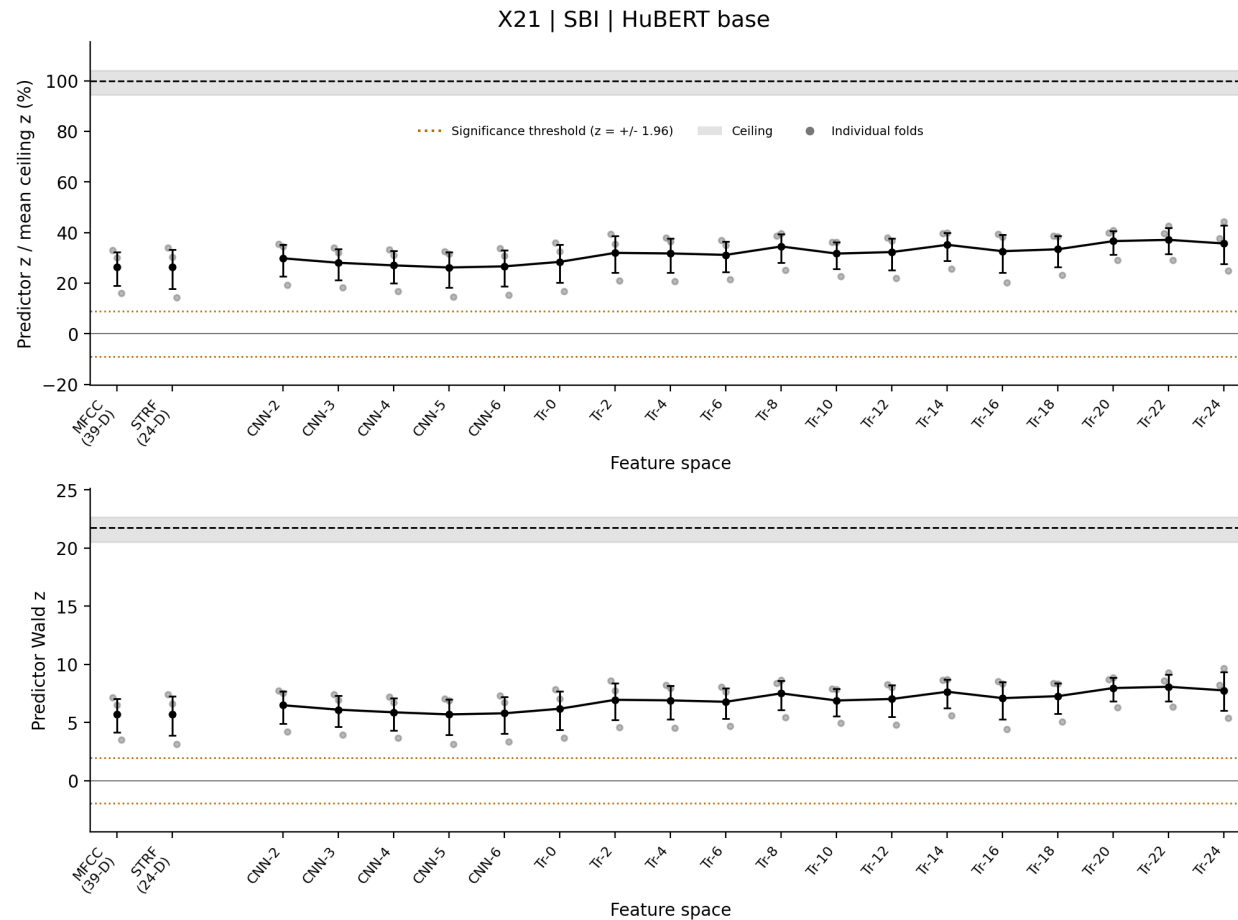


This shows **control-test performance**, not exposure performance. I added marginal distributions, a shared legend and dataset-only titles using your map colors. Each marginal distribution counts a matched test-word recording once, irrespective of listener count.

This conceptually extends [Kim et al. \(2025\)](#), whose L2-to-English distances predicted relative talker intelligibility. Our matched-word, control-test analysis differs in design and unit of analysis.

Curves retain 95% intervals from 1,000 participant bootstraps: AN19 has 1,880 responses; X21 has 4,117. The 40 AN19 wave/wade responses remain excluded pending audio verification. The first-ten-exposure analysis remains separate, with the window defined before examining results.

SBI: z across feature spaces



I reduced the figure width, shortened the x-axis label to "Feature space", and moved a three-entry legend below the ceiling. Points show individual folds and their mean with a 95% fold-bootstrap interval. The upper panel divides z by the mean behavioral ceiling; the lower panel retains raw z . The gray band shows ceiling uncertainty.

These remain the earlier notebook test-fold-refit statistics. The orange references are nominal $z = \pm 1.96$; I have retained those per your annotation. They are not multiplicity-adjusted tests of the mean curve, and normalized z is not percentage variance explained. The same revised layout is available for all three datasets and both model variants.

SBI: the likelihood companion

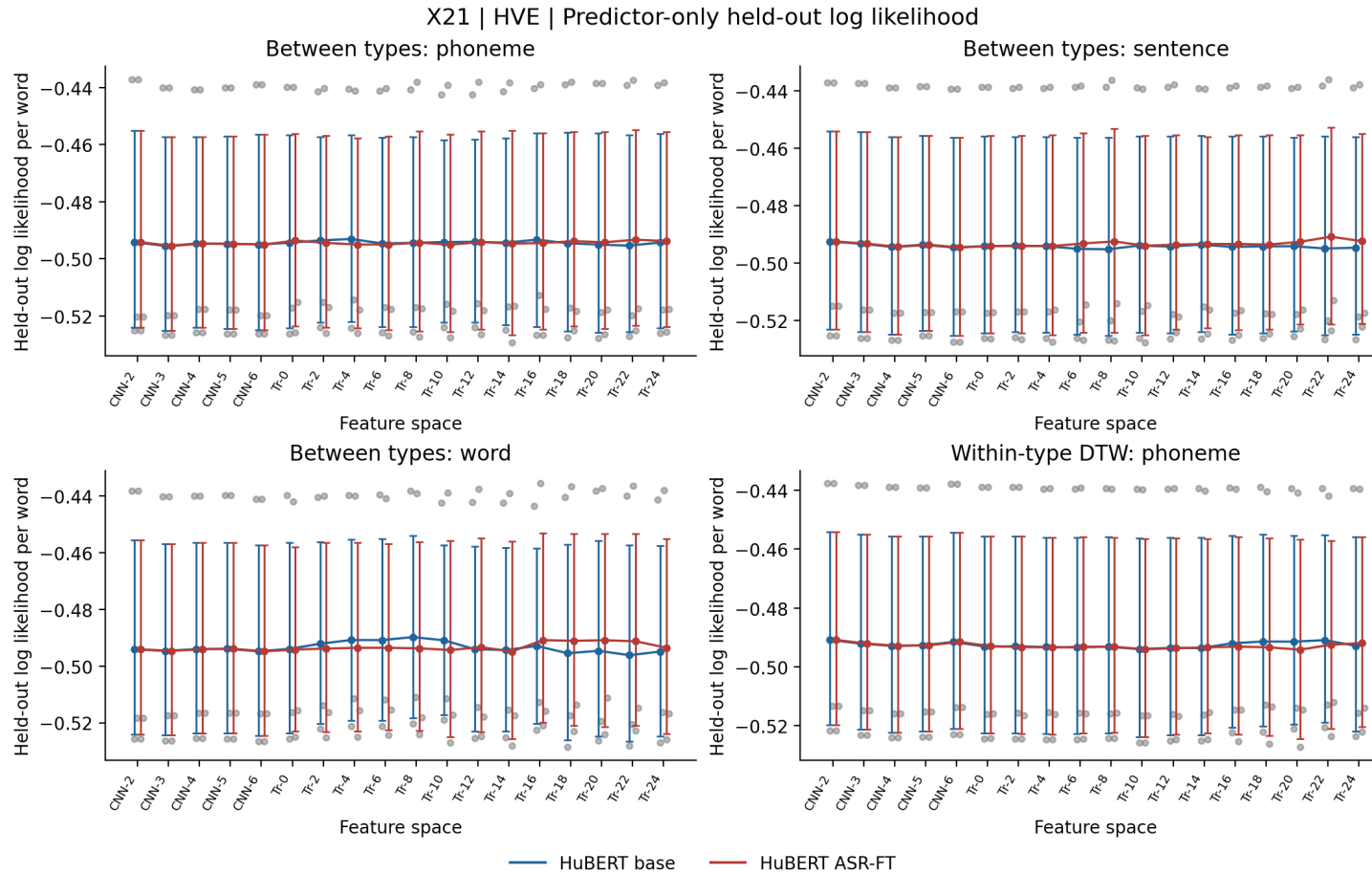


These are recovered predictor-only held-out scores for every registered layer and both acoustic baselines, using matching response rows. Higher is better. Log likelihood per word accounts for fold size; points, means and 95% bootstrap intervals use the three folds.

These runs used training-standardized negative DTW, whereas the preceding notebook z plot used exponential similarity. This is therefore not a matched comparison of optimization objectives; that requires the same predictor construction and sample.

The ceiling line awaits the item-by-condition reference: current AN19/X21 keys omit condition. AN19 also varies its talker random-intercept structure after fitting fallbacks; a matched-structure rerun is needed for a controlled layer comparison.

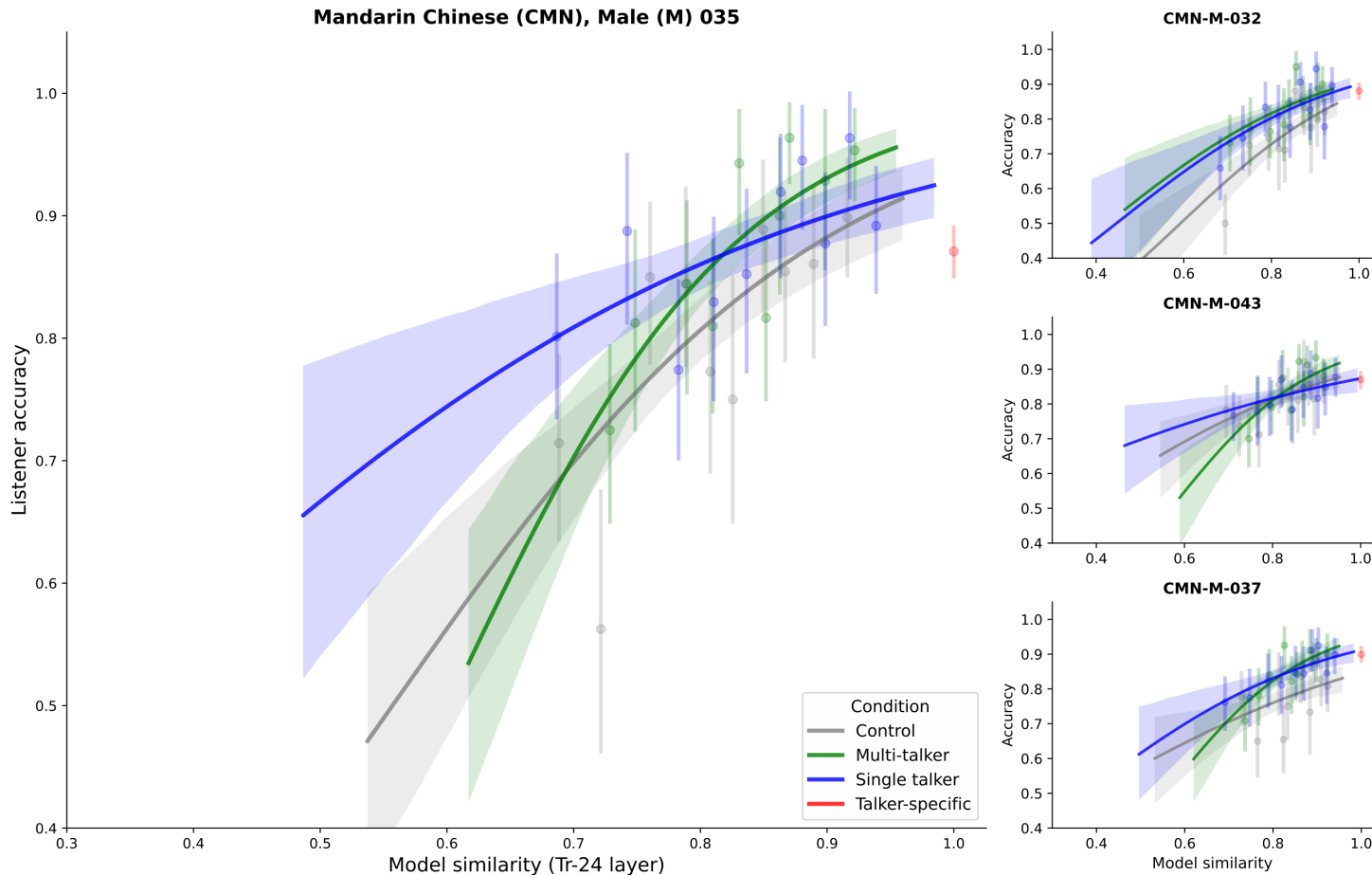
HVE: likelihood across layers and definitions



I also added likelihood profiles for every available revised HVE definition, across all 18 registered layers and both HuBERT variants. This page shows four X21 definitions; the accompanying figures contain the remaining definitions and datasets. All curves show predictor-only held-out scores with 95% three-fold bootstrap intervals.

For B23, I keep the 97-participant global-order sample separate from the 168-participant order-independent sample. I do not rank those samples against each other by total log likelihood. The revised HVE z profiles remain available separately; the old notebook ceiling should not normalize their training-fold z values.

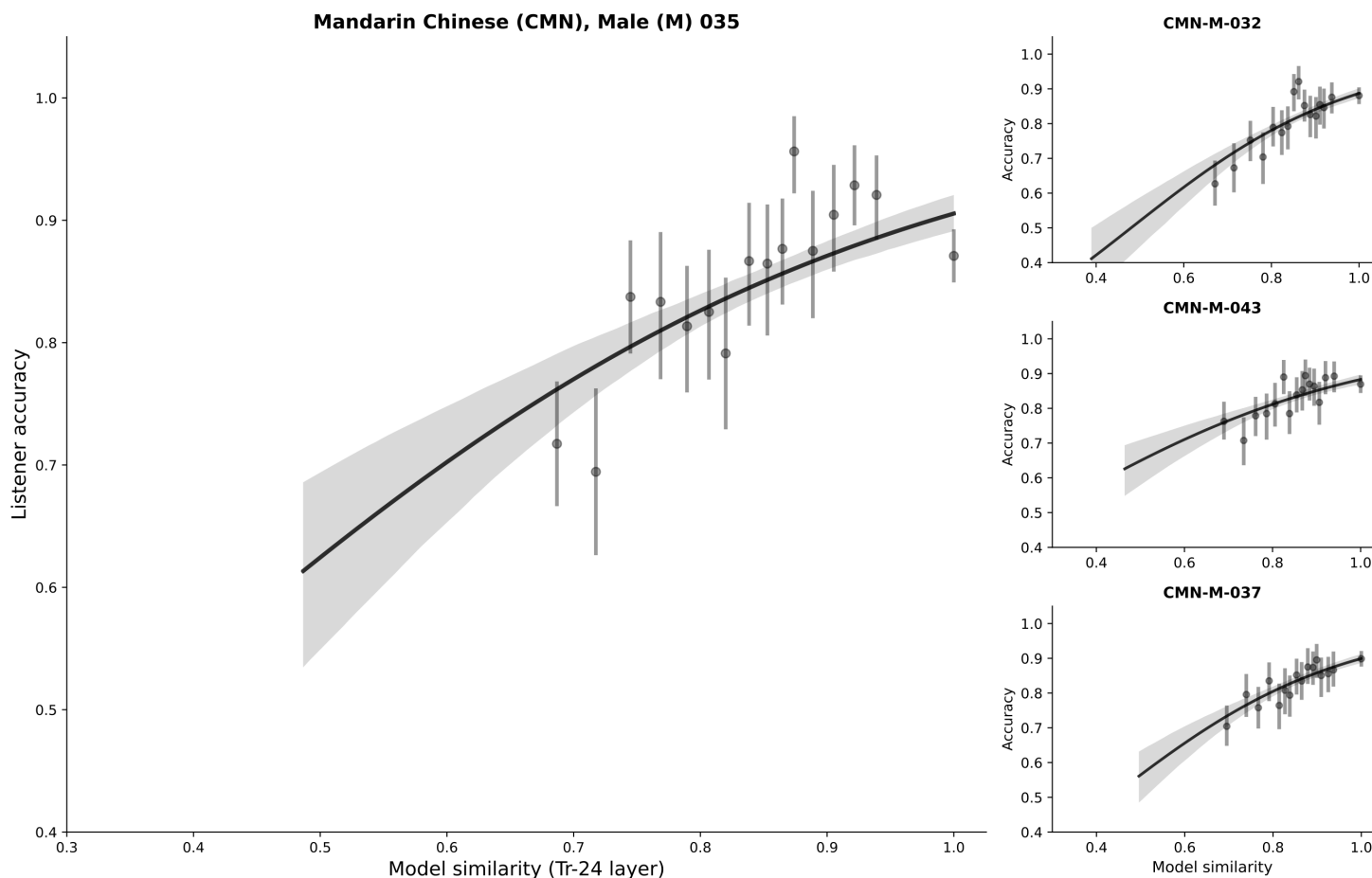
X21: condition-specific similarity curves



I moved the language ahead of the talker identifier and corrected CMN in the titles. The main panel is talker 035; the smaller panels are 032, 043 and 037. These plots use exposure-test similarity rather than the English-reference similarity in Figure 2c.

I retained the original descriptive curves, points and intervals. Their existing metadata do not identify the bootstrap resampling unit, so I have not relabeled these intervals as participant-bootstrap or three-fold intervals. The model comparisons, rather than the curve offsets alone, test the additional contribution of condition.

X21: pooled across conditions



This is the requested single black curve and point series for each talker. I pooled the trial rows across conditions, using 20 bins instead of the 10 per-condition bins in the preceding figure. It is not an equal-weight average of four separately fitted condition curves.

Both versions use the same 16,477 trial responses from 320 participants. I have retained the existing fits and uncertainty intervals rather than refitting the human-response model for this display update.

Clarifying the X21 HVE and condition comparisons

I checked the original tables: SBI and HVE use the same 16,477 responses, 320 participants, 412 items and four test talkers in X21. Their condition-only models have identical log likelihood, -5484.3114.

Quantity	HVE: CNN-6 within-word transitions	SBI: base Tr-14
Predictor-only z	-5.5038 (p = 3.72e-8)	4.7807
Predictor z after adding condition	-1.9516	2.6336
Predictor beyond condition	Chi-square(1) = 3.7383; p = .05318	Chi-square(1) = 6.5176; p = .01068
Condition beyond predictor	Chi-square(3) = 11.0085; p = .01168	Chi-square(3) = 21.0030; p = .000105

The borderline HVE result refers to its additional contribution **after condition is included**. HVE has a strong negative association in the predictor-only model, and condition still adds significantly to it. The negative direction is not support for the expected positive variability effect.

Condition adds less likelihood to HVE than to SBI here because the two reduced models already capture different parts of the responses. These incremental tests need not be equal. The selected HVE base/FT inputs and results are identical, so I do not count them as two independent replications. These combined-fold tests remain exploratory because layer/method selection used the same study.

A shared ceiling and comparable analysis samples

My earlier phrase "an HVE-compatible ceiling" was confusing. The historical SBI and HVE ceiling files already contain exactly the same nine fold values. I should use one behavioral reference for the two theories when the response rows, folds, statistic and fitting procedure are the same.

I do still need to resolve two practical differences. The historical ceiling is a test-fold-refit z, whereas revised HVE coefficients come from training-fold fits. Also, the direct AN19/X21 prediction ceiling currently groups by item and talker without condition; I need the item-by-condition grouping we discussed.

Dataset	SBI sample	HVE sample
AN19	120 participants; 5,685 word responses	120 participants; 5,760 word responses
X21	320 participants; 16,477 word responses	The same responses and model structure
B23	168 participants; 10,080 count rows	Same size for order-independent HVE; 97 participants and 5,820 rows for global order

In AN19, the 75 extra HVE responses involve hung, mote, main and route. The selected SBI fit also dropped the talker random intercept after fitting problems, while HVE retained it. I need to align both the rows and random-effects structure before directly comparing the two theories there. Each existing within-family nested comparison uses its own matched rows and structure.

What the acoustic scaling mismatch means

The full MFCC/STRF analysis standardizes each acoustic coordinate before calculating DTW: x becomes $(x - \text{corpus mean}) / \text{corpus standard deviation}$. The component analysis used the original coordinate values. I verified that this difference occurs across the same 5,459 recording pairs.

This changes how heavily each acoustic dimension contributes to distance and can change the DTW alignment. It is not the later standardization of the single GLMM predictor, and standardizing that predictor cannot undo it.

To identify which acoustic components account for AN19's strong result, I need to rerun the component subsets using the corresponding coordinates of the full baseline's standardizer, the same recordings, DTW settings, folds and response rows. Until then, I cannot interpret the component ranking as the effect of removing those dimensions. This mismatch alone does not show that the full-baseline result is wrong.

Next analyses and materials

1. **Match the inferential comparisons.** Align response samples and random-effects structures; use a shared, item-by-condition behavioral reference for a defined statistic and evaluation procedure.
2. **Rerun the acoustic components.** Apply the same coordinate scaling and compare both z and likelihood on matched data.
3. **Pilot phoneme alignment.** AN19 lacks existing labels, but I can assess automatic alignment against known target words, then manually check a small, prespecified sample. ASR output alone would not establish reliable phone boundaries. Alternatively, I can use a dataset with existing annotations, while retaining the limitation that it covers fewer L1 backgrounds. The lexical-error analysis additionally needs response-to-phone alignment and a pronunciation lexicon.
4. **Add early-exposure curves.** Define the first ten presentations and their associated word responses before examining the result, keep them separate from control-test performance, and check which datasets have usable trial order and responses.
5. **Finish Figure 1 sources.** Check tone coverage and weighting, obtain the feature-heatmap score table/code, and assemble the inventory summary above the examples. I have kept your original map and code alongside the revised figures.

The figures are saved locally in PNG, SVG and PDF where generated from code. I have not committed them to Git or uploaded them to Overleaf. The detailed comment-by-comment record and outstanding items are in REVIEW_RESPONSE.md and TODO.